

Authority Delta

**Human decisions that remain accountable
as evidence and AI capabilities change**

Users, workflow, service boundaries and verified behavior

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Who Authority Delta helps

Person	Work that creates friction	What changes with ReadinessOps
Operations lead / AI program owner	Reread changed evidence and reconstruct the basis of an earlier decision.	A cited candidate compares changes with the exact published decision.
Action owner	Track whether completing a task actually resolves the original issue.	Completion requires new resolution evidence for subsequent reassessment.
AI platform developer	Connect agent capability to explicit human authority and a reliable stop path.	Separate approval, publication and finite AWS delegation with verified removal.
Reviewer / assurance team	Reconstruct who decided what and which evidence supported it.	Versioned records preserve proposals, human decisions and operational history.

Intended benefit: less repeated review preparation. Productivity and commercial savings remain unmeasured.

An operations lead needs to know whether an issue is resolved

THE PERSON'S QUESTION

"Can we reconsider the earlier decision?"

An earlier action asked for evidence that execution authority could be stopped.

The action owner supplies new evidence

The owner records completion and attaches a new verification document.

The lead still owns the judgment

The lead reviews the proposed changes before deciding whether to publish.

OBSERVED PRODUCT BEHAVIOR

A fixed reassessment input

The assessment selects the resolution evidence and original acceptance description.

A specific change in the candidate

Governance and Portfolio are affected. Value and Model routing remain unknown.

The official decision remains intact

The later candidate is Ready for review. Existing publication and authority stay unchanged.

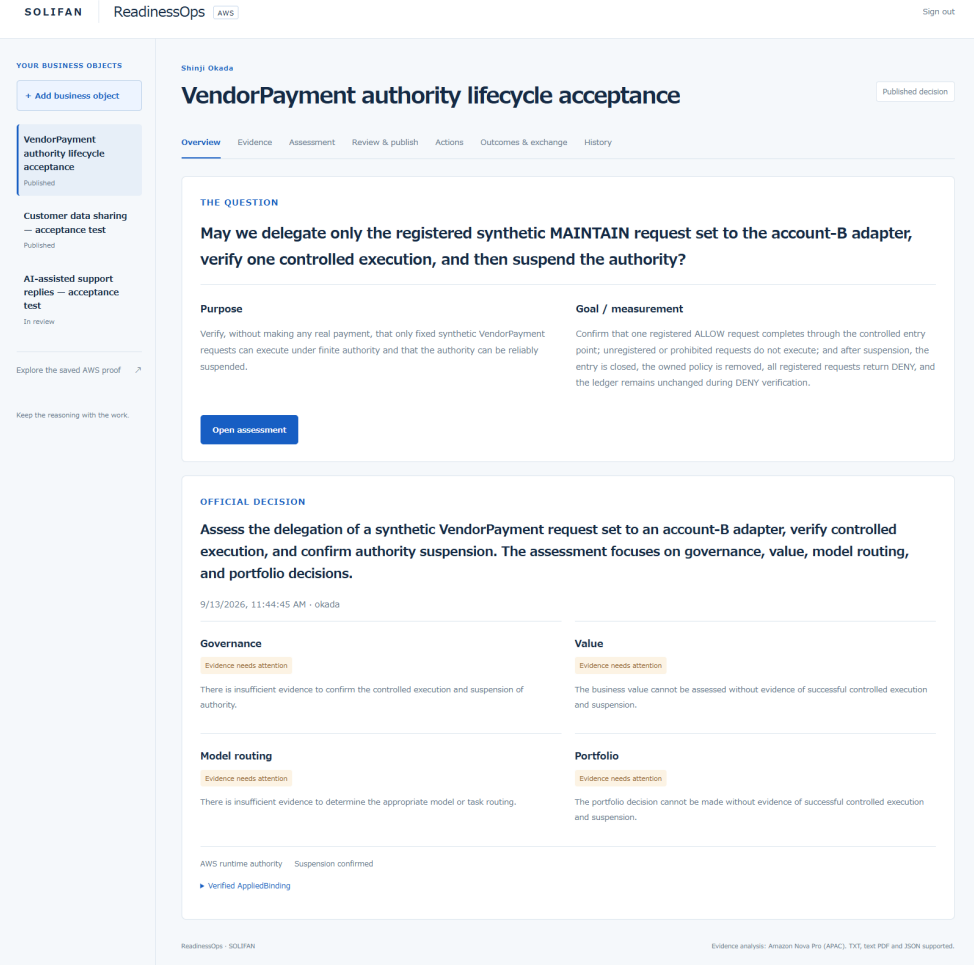
Observed RO-08 scenario: new evidence changes the review candidate without renewing execution permission.

One business record connects judgment to follow-up

Stage	Person’s responsibility	Product responsibility
Question and evidence	Define the question, owner, goals and measures. Select evidence.	Retain original files, extracted text and versioned evidence.
Assessment and review	Start assessment. Check citations, edit and record judgment.	Prepare cited findings, proposed actions and decision items.
Approval and publication	Approve a specific revision, then explicitly publish it.	Record approval separately from the official publication.
Actions and outcomes	Assign ownership and due dates. Supply resolution evidence and measured results.	Retain completion evidence, outcome provenance and history.
Reassessment	Review the impact of new evidence, rules or an agent release.	Compare with the pinned publication and keep unknowns explicit.

The complete core workflow works without an execution adapter or a second AWS account.

The reviewer sees the question beside the official decision



The business question provides context

A named business object holds its purpose, success measures and the published decision.

Seven views retain the working record

Evidence, assessment, review, actions and outcomes stay connected to the same object and its history.

Publication and AWS authority remain distinct

Retained repository capture. The later RO-08 review candidate does not replace this official record.

Authority Delta focuses the next human review

Comparison input: the selected evidence snapshot plus the exact prior publication and stable Decision Item identities.

Perspective	Observed impact	Interpretation for the reviewer
Governance	Affected	Controlled execution and confirmed suspension change the basis for review.
Value	Unknown	The technical acceptance does not establish commercial value.
Model routing	Unknown	The selected evidence does not establish model or task-routing suitability.
Portfolio	Affected	The synthetic initiative can be reconsidered within its demonstrated scope.

The contract supports Affected, Unchanged and Unknown. The observed RO-08 run had no Unchanged item.

A valid quotation proves source fidelity. A human must still judge relevance and interpretation.

Each consequential transition has a separate record

Record	Meaning	Developer constraint
Assessment / candidate	AI-generated findings and proposed decisions.	Cannot approve, publish or change AWS Policy.
ApprovalReceipt	Authenticated human judgment on a revision.	Approval alone does not replace the official publication.
DecisionPublication / PublishedCurrent	The explicitly published official business decision.	Publication alone grants no runtime authority.
AWS delegation receipt	Separate human approval of exact, finite connected authority.	Bind publication, connection, release, requests, Policy hash, expiry and generation.
AppliedBinding / execution result	Verified applied authority and a retained invocation outcome.	Require verified application evidence and current authority checks.

New evidence and imported references cannot silently promote a proposal into official authority.

AWS services support a recoverable evidence-to-decision workflow

Layer	AWS implementation	Responsibility and boundary
Workspace and identity	CloudFront, React/TypeScript, Cognito	People issue business commands. The browser does not supply trusted Policy text.
Authenticated command API	API Gateway and Lambda	Verify JWT subject/scopes. Check revisions and request IDs conditionally.
Assessment processing	SQS, Strands Agents SDK, AgentCore Runtime, Nova Pro	Read fixed inputs and generate cited proposals. No approval or Policy mutation.
Versioned record	DynamoDB and versioned S3	Keep evidence, snapshots, runs, drafts, receipts and publications version-fixed.
Recovery	DynamoDB outboxes, SQS/DLQ, scheduled dispatch	Recover duplicate delivery, lost responses and uncertain operations.

CodeBuild supports deployment and readback. Ordinary user assessments run through the queued service path.

Connected authority has separate deployment and execution paths

Purpose	ReadinessOps origin	Registered target
Deployment verification	CodeBuild assumes read-only DiscoveryRole.	Read back Runtime, Gateway, Policy Engine, functions and immutable registry.
Apply or revoke Policy	Application/Revocation Worker assumes PublishInvokeRole.	Qualified Limited Publisher Lambda owns the exact Policy lifecycle and proof.
Normal invocation	Invocation Gate accepts current authority. Worker assumes RuntimeInvokeRole.	Qualified Invocation Lambda checks the active journal, then calls Runtime and Gateway/Policy.

The verified connector uses distinct accounts and ExternalId-bound roles. The normal invocation path bypasses the Policy Publisher.

VendorPayment is a synthetic adapter. The supplied connector requires a distinct target account.

Suspension requires proof after entry closes

Sequence	Required behavior	Why the developer must preserve it
1 Close entry	Invalidate the current delegation generation.	New invocations cannot enter the closed generation.
2 Drain accepted work	Resolve already accepted invocations before removal.	Stop and invocation acceptance remain consistently ordered.
3 Remove exact authority	Remove only the Policy owned by this application.	Revocation cannot expand into unrelated customer permissions.
4 Verify live denial	Every registered request returns fresh DENY and the ledger stays unchanged.	A deletion response alone cannot establish suspension.
5 Confirm matching proof	Match the current application, enforcement digest and revoke request.	Only complete matching evidence reaches SUSPENDED_CONFIRMED.

UNKNOWN remains nonterminal. Lost responses retain the writer fence and reconcile the same operation.

Developers can extend the workflow without widening AI authority

Development goal	Primary code boundary	Invariant to preserve
Improve assessment quality	services/business_analysis/ src/authority_delta/business/	Use fixed inputs, cited output and stable prior-item identities. Keep invalid runs visible.
Add a business workflow	Business contracts, API and workbench/	Preserve explicit human judgment, publication, ownership and resolution evidence.
Integrate another execution target	Adapter contracts and services/customer_publisher/	Register finite requests, separate delegation and implement verifiable removal.
Improve operational resilience	services/business/ and infra/	Preserve conditional writes, idempotency, outbox recovery and uncertain-state handling.

The non-payment common workflow exists. Another live execution adapter requires its own implementation and acceptance evidence.

Observed AWS results establish specific behaviors

Evidence	Observed result	Limit of the claim
RO-07 Authority lifecycle	One registered ALLOW, exact Policy removal and seven registered DENY results after suspension.	Synthetic VendorPayment only. No real payment or unrestricted execution.
RO-08 Action and reassessment	New resolution evidence produced a validated candidate ready for review on runtime 25.	One observed scenario. Earlier failed runs remain. No new publication or delegation.
RO-10 Outcomes	Two outcomes distinguish observed technical proof from unmeasured commercial value.	Commercial savings and productivity improvement remain unmeasured.
RO-11 Reference exchange	Import and repeat import retained one fixed reference without changing official records.	File-based reference only. No live Snowflake connection.

Local tests and offline verification support the evidence. They do not replace authenticated live acceptance.

A reproducible local starting point

Python 3.12 on Linux x86_64, plus Node.js and npm. Run these commands from the repository root.

Step	Command
Environment	<code>python3.12 -m venv .venv-analysis</code>
Pinned dependencies	<code>.venv-analysis/bin/python -m pip install --require-hashes -r requirements-analysis-x86_64.lock</code>
Frontend dependencies	<code>npm ci --prefix workbench</code>
Local verification	<code>PYTHONPATH=src:scripts:tests:. .venv-analysis/bin/python -m unittest discover -s tests -q</code>
Frontend build	<code>npm run build --prefix workbench</code>

Commands shown over two lines are single shell commands. Fresh checkouts may not contain distribution bundles or prebuilt environments.

The next file depends on the problem you are solving

Question	Repository entry point
How do users complete the workflow?	README.md and submission/JUDGE_GUIDE.md
Where do service and authority boundaries sit?	docs/ARCHITECTURE.md and docs/IMPLEMENTATION.md
How do actions feed the next decision?	docs/BUSINESS_ACTION_LIFECYCLE.md and docs/RO08_LIVE_ACCEPTANCE.md
How do I deploy or recover an existing workspace?	docs/DEVELOPMENT.md and docs/READINESSOPS_WORKSPACE_UPDATE.md
Where is the proof for applied and removed authority?	docs/RO07_LIVE_ACCEPTANCE.md and evidence/

Repository snapshot solidfdn/readinessops-authority-delta, commit 687b71e2553c

<https://github.com/solidfdn/readinessops-authority-delta>

A useful first implementation has an accountable human owner

FIRST USE CASE

One consequential business question

Choose a recurring review where new evidence can change a published decision.

An owner and a reviewer

Assign responsibility for follow-up work and for the official judgment.

A measure of usefulness

Measure review preparation effort and whether resolution evidence answers the question.

ENGINEERING ACCEPTANCE

The core record stays complete

Preserve originals, failed runs and decision history while candidates evolve.

Human transitions stay explicit

Keep approval, publication and connected delegation as separate commands.

Connected authority stays finite

Add execution only with registered scope, expiry and verified removal.

Authority Delta helps people reconsider decisions with evidence while retaining responsibility for what happens next.